

Emotion-Based Human–Computer Interaction

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Abstract

Emotion-aware systems provide the capability of making HCI much more natural and adaptive insofar as the system responds to a user's affective state. This project proposes an end-to-end pipeline for FER to classify eight categories of emotions including anger, disgust, fear, happiness, sadness, surprise, neutral, and contempt using a combined dataset derived from FER-2013 and CK+. The system adopts face cropping with CLAHE-based contrast enhancement, a baseline CNN, attention-enhanced CBAM-CNN, and a transfer-learning model based on EfficientNet-B0. This treats significant class imbalance among these emotions with a weighted random sampler at the time of training.

It also reports the performance of the models in terms of overall accuracy, per-class metrics, ROC curves, and gradient-based saliency visualizations for interpretability. Compared to the test accuracies of circa 47% and 56% obtained by the baseline CNN and CBAM-CNN, respectively, the best performing EfficientNet-B0 reaches a peak performance of 66.49% on the combined test set. These results give evidence that happiness and neutral expression are very well detected while fear and disgust are still rather difficult to be identified due to subtle facial cues and scarce representation in the dataset.

Overall, the system demonstrates promising robustness for practical emotion-aware HCI applications, such as adaptive learning dashboards, mental-wellness tracking, and responsive UI. The work concludes by suggesting the way ahead to improve fairness, generalization, and temporal modeling in future FER systems. Besides, the saliency maps provided transparent insight into model decision-making processes, quite a critical factor when any FER system is deployed in sensitive HCI environments. The combined FER-2013+CK+ training strategy further demonstrates how such diverse datasets lead to improved real-world robustness and hence make the system more practical for user-facing applications.

Introduction

Human–Computer Interaction has traditionally assumed that users approach systems in some sort of neutral state; the computer responds only to explicit inputs, whether by click, keystroke, or gesture. At the same time, there is a great number of applications for which real-world effectiveness can be seriously enhanced if systems were able to sense-and-respond to the user's emotional state, such as remote learning, mental health support, gaming, customer service, and assistive technologies. Emotion-aware HCI enhances personalization, engagement, and user experience in general.

Facial expression recognition perhaps represents one of the most viable and non-intrusive approaches for detecting human feelings. The human face conveys rich emotional information in an immediate manner, where recently deep learning architectures have made automatic FER non-intrusive and enormously scalable. On the other hand, benchmark datasets like FER-2013 consist of in-the-wild images, whereas CK+ contains posed high-quality sequences. These have made them extremely popular in the evaluation of FER systems.

In this context, the project develops an end-to-end FER pipeline that should provide newer dimensions to Emotion-Based HCI. We have trained and compared a baseline CNN, CBAM-enhanced CNN, and transfer-learning model based on EfficientNet-B0 in an aggregated FER-2013 + CK+ dataset. It also incorporates face detection, CLAHE-based preprocessing, class balancing, and multiple evaluation metrics.

Beyond classification accuracy, this project strongly emphasizes interpretability through confusion matrices, ROC curves, and gradient-based saliency maps that let us understand which facial regions drive model decisions. Finally, we present a discussion on how such FER models may support real-world HCI applications of adaptive interfaces, stress monitoring, and learning support systems.

Related Work

3.1 Classical and Early Deep FER

Facial expression recognition was pegged as a deep learning benchmark with the contribution of the FER-2013 dataset, which was introduced as part of the ICML challenge on representation learning which has difficult real-world samples [1]. Likewise, Extended Cohn Kanade (CK+) extended the other CK data by providing reliable emotion labels and action unit annotations, and furthermore, it is also one of the most popular posed expression datasets [2]. AffectNet subsequently developed the state of the art by making more than a million annotated images available with categorical emotions and continuous valence-arousal ratings, facilitating a large-scale FER in-the-wild study [3]. These three datasets comprise FER-2013, CK+, and AffectNet and make the history of contemporary FER research.

3.2 FER Data Augmentation and Deep CNN.

There is a large amount of literature that addresses CNN-based FER architectures and data augmentation methods intended to enhance noise or imbalance resilience in datasets. Georgescu et al. proved that using deep CNN features in conjunction with handcrafted bag-of-visual-words representations can achieve more than 75% accuracy on FER-2013 when using local learning and SVMs [7]. More modern hybrid and ensemble CNN models all tend to reach 65-75 percent accuracy with deeper networks and strong augmentation [8]. Yalcin and Alisawi demonstrated that properly resolving mislabeled, and poor-quality FER-2013 samples help dramatically improve deep models, and therefore, preprocessing is a crucial component of FER pipelines [6]. This is the reason we have been encouraged to focus on face cropping, CLAHE refinement and the handling of our datasets.

3.3 Dynamic Attentional Processes and Elegant Architectures

The architecture of attention has gained prominence in FER since it can highlight expressive regions in faces. An example of these methods is CBAM, which repeatedly uses channel and spatial attention to sharpen CNN feature maps and has been demonstrated to be able to enhance performance on a range of vision tasks [4]. In contrast, EfficientNet uses neural architecture search and compound scaling to generate light models with high accuracy CNNs, so it can be applied to real-time

FER and edge-device HCI systems [5]. These are the developments that guide our selection of CBAM-CNN and EfficientNet-B0 as fundamental structures to make the comparisons.

3.4 Emotion-Based HCI

Emotion recognition is also at the center of intelligent user interfaces and affective computing systems. Recent surveys speak about the role that FER plays in adaptive learning platforms, mental health monitoring, available communication technologies, and real-time user experience assessment [9]. FER methods that utilize deep learning are also under investigation to be applied in the real world, such as action-unit-aware networks [10]. Our research is in line with this direction as it explicitly orientates FER as the way of enhancing adaptive and emotion awarding HCI performance.

Recent FER studies are putting more emphasis on the robustness and real-world flexibility as emphasized in emotion categorization surveys and database design surveys [9]. In addition to classical CNNs, state-of-the-art models are based on attention (like CBAM [4]) and compound-scaled models (like EfficientNet [5]) to obtain enhanced generalization in unrestricted settings. It has also been found that quality of preprocessing and reliability of data have a big influence on performance, which supports the results of Yalcin and Alisawi [6] and Georgescu et al. [7]. More modern hybrid and real-time FER models also put a stronger emphasis on efficiency and deployment preparedness of HCI situations [8]. FER is also being bridged to emerging work that better interprets emotion-sensitive systems with the assistance of deeper representations of facial action units [10]. These advances are in line with our project use of refined preprocessing, attention modules, and transfer learning to be able to support adaptive HCI applications.

Problem Definition

The traditional HCI systems do not take cognizance of the emotional state of the user and therefore, they do not have the ability to adapt, respond and personalize human interaction effectively. In case of using application like e-learning, health sector support, customer service, and mental-wellness monitoring, identifying the emotions of the users will be very important in developing intelligent, intuitive and empathetic interfaces. Facial Expression Recognition (FER) offers a mildly invasive, generalizable approach to drawing an emotional conclusion based on visual cues, yet real-world reliability performance is difficult to attain.

The issue that is solved in this project is to create and test a powerful FER pipeline that can categorize various emotions states based on deep learning. In particular, the system should be capable of successfully identifying eight categories of emotions, including anger, disgust, fear, happiness, sadness, surprise, neutral, and contempt, made of combining and preprocessing two popular datasets, FER-2013 and CK+. Issues such as the imbalance of datasets, noise and low-quality image management, balancing in-the-wild and posed expression differences and the consistency of labeling across datasets are all problems.

Moreover, the project aims at examining the effect of preprocessing (face-cropping and CLAHE) methodologies, attention (CBAM), and transfer learning (EfficientNet-B0) on performance. Another objective is to determine the architectural decisions that provide optimal generalization and interpretability, and to find out whether the concluding model can be used as a sound base to support adaptive, emotion-sensitive HCI applications.

In the end, not only the categorization of emotions in the most accurate way is required, but the creation of the FER system is also supposed to be scalable, interpretable, and applicable to real-time, user-focused settings.

Dataset Description

5.1 FER-2013

Even the most common FER benchmark is the FER-2013 which consists of 48x48 grayscale images of facial expressions. The given original dataset consists of 35,887 images, as based on which seven emotions were classified (anger, disgust, fear, happy, sad, surprise, neutral) [1]. Kaggle is rearranged with a train and test folder.

5.2 CK+ (Extended Cohn–Kanade)

CK+ has 593 image sequences of 123 subjects whose FACS and emotion labels are valid and encompass eight emotions such as contempt [2]. We use a pre-split (ck+split) version (train/ test folders), where there is an emotion label in each folder.

5.3 Combined FER+CK+ Dataset

We regard the FER-2013 and CK+ prepared splits as combined by:

- Combining their train folders to come up with a single PRETRAIN.
- combining the folders testing them with PRETEST.

Image Folder Automatic mapping of class names to integer labels.

This generates eight classes of emotions:



Figure 1. Example preprocessed and augmented face images from the combined FER-2013 + CK+ training set.

1. angry
2. disgust
3. fear
4. happy
5. neutral
6. sad
7. surprise
8. contempt

The combined dataset of FER-2013 and CK+ is inherently imbalanced in classes, with happy and neutral emotions featuring much more than disgust or contempt. That would present a model biased toward the prediction of majority classes and make the learning of minority emotions far more difficult. Weighted sampling hence forms the basis of training to ensure each class proportionally contributes to the learning process so that underrepresented categories are not ignored by the model.

Merging the two datasets brings great visual diversity, since the FER-2013 contains noisy, low-resolution, and in-the-wild images, while the CK+ images are cleaner, well-framed, and posed expressions. This heterogeneity thus forms a more challenging but realistic environment for training the models, both on subtle micro-expressions and on more exaggerated facial cues. With that said, this network increases its ability to generalize to real users interacting under varied lighting, pose, and expression intensity outside of the controlled laboratory environment.

Since there are differences in contrast, clarity, and framing of faces across these datasets, some preprocessing steps should be used to regularize the visual features before training, such as face cropping and CLAHE. These steps can help establish more coherent input representations and reinforce the reliability of model comparisons across CNN, CBAM-CNN, and EfficientNet architectures. Therefore, a unified and well-preprocessed dataset can provide a solid basis for developing FER models competent for real-world emotion-aware HCI applications.

Methodology and Implementation

6.1 Overall System Pipeline

The FER- HCI system adheres to an entire end-to-end pipeline beginning with the storage of crude images up to the ultimate emotion forecasting. The major stages are:

1.Dataset Loading

Both the FER-2013 and CK+ directory structures are loaded through Google drive. CK+ sequences are converted into single-frame samples to be in FER-2013 format.

2.Face Detection and Preprocessing

The localization of the face area is done in all images via OpenCV Haarcascade_frontalface_default.xml.

- In case there is more than one face, then the biggest bounding box is picked.
- Cropped faces are resized (CNN/CBAM: **180×180**, EfficientNet-B0: **224×224**).
- CLAHE is used to improve the local contrast and minimize illumination noise.

3.Data Integration

Fer-2013 and CK+ processed images are combined to obtain a single set of FER+CK+ dataset.

WeightedRandomSampler corrects class imbalance and makes the rarer classes (e.g., disgust, contempt) equally participated in training.

4.Model Training

Three architectures are trained to compare:

- Simple CNN (Baseline)
- CBAM-CNN channel + spatial attention.
- EfficientNet-B0, partially frozen, ImageNet pretraining.

5.Measurability & Interpretability.

The performance is measured with the help of:

- Class accuracy
- Confusion matrix
- ROC–AUC curves
- The visualization of saliency-maps identifying regions of the face as discriminative.

6.2 Integration of Preprocessing Pipeline and Dataset

Consistency across the FER-2013 and CK+ datasets, which are very different in terms of image quality, labeling formats, and expression properties, can be achieved only through the preprocessing stage. The data sets are stored on Google Drive and face detection of each sample is done with Haarcascade frontal face classifier. When there is more than one face in an image, the biggest bounding box will be chosen to make sure that the model pays attention to the focus. The face is then identified, cropped, resized and improved with CLAHE to obtain a better face, with the local contrast made more pronounced, and the minor features that represent the expression. After the preprocessing is done the clean images of both datasets are again rearranged under common train and test directories and can be easily integrated with the PyTorch Image Folder utility. To reduce class imbalance in the combined dataset (which is especially high in the minority emotions, e.g., disgust and contempt), a WeightedRandomSampler is used. This would make sure that the underrepresented classes would not be less represented during training so that the models would not be biased to the prevalent types of emotions.

6.3 Data Augmentation and Data Normalization

To minimize overfitting and enhance generalization, multiple augmentation strategies are used in training. There are also random flipping and rotation of images over a limited range to create a realistic simulation of changes in facial positioning. The next step that follows augmentation is normalization which is used to rescales the intensities of pixel values to a normalized distribution, to enable the gradients to be stable when updating. CNN and CBAM-CNN models can be used with 180x180 grayscale images, while EfficientNet-B0 can be used with 3-channel input images, so the grayscale images must be converted into 3 channels before being resized to 224x224. All these preprocessing steps are useful in enabling the models to learn robust features in various visual conditions.

To make it more robust and less prone to overfitting, the augmentation operations are augmented by the following:

- Random horizontal flip
 - Random rotation ($\pm 10^\circ$)
 - Resize
1. CNN / CBAM: 180×180
 2. EffNet-B0: 224×224
- Normalize to mean 0.5, std 0.5

This makes sure that models are subjected to different face orientations and light settings when training.

6.4 Baseline CNN

The Baseline CNN is the architecture that the other, more advanced models are compared to. It is composed of four convolutional blocks that include a 3×3 convolution, ReLU activation, and max pooling which are used in increasing numbers of 32, 256 feature channels. Once the final convolutional layer has been performed, global average pooling is performed, and then the final layer is a fully connected layer, which classifies the facial expression into one of the 7 categories of emotions. The model is trained with 40 epochs of cross-entropy loss and Adam optimizer and a batch size of 32. This simple baseline architecture is however a useful lower bound in assessing the gains made by attention processes and transfer learning.

Block	Conv Channels	Operations
1	$3 \rightarrow 32$	Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool
2	$32 \rightarrow 64$	Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool
3	$64 \rightarrow 128$	Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool
4	$128 \rightarrow 256$	Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool

Table 1. CNN block architecture

6.5 CBAM-CNN

The CBAM-CNN is an improvement of the baseline model to boost its representational power, adding the Convolutional Block Attention Module (CBAM) on top of each convolutional block. CBAM focuses the feature maps in sequential channel and spatial attention to enable the model to highlight feature maps with the more informative information and facial regions. The channel attention module determines important channels of features using max- and avg-pooling mechanisms, and the spatial attention module illustrates important areas of the pixel that are important regarding expression cues. Upon exiting four attention-enhanced blocks, the network downsizes the $256 \times 11 \times 11$ feature map and then runs it through a 512-unit fully connected layer including dropout; it is then finally classified. The CBAM-CNN, trained under the same conditions as the baseline CNN, has better recognition results, which proves that attention is effective in low-resolution FER tasks.

The CBAM-CNN is an extension of the baseline CNN by adding CBAM block following every convolutional module.

CBAM Components

1.Channel Attention (CAM)

- AvgPool + MaxPool $\rightarrow$ Shared MLP $\rightarrow$ Sigmoid
- Gets to know which of the channels are important.

2.Spatial Attention (SAM)

- AvgPool (Channel-wise) + MaxPool (Channel-wise) $\rightarrow 7 \times 7$ Conv $\rightarrow$ Sigmoid
- Flexible localizations of crucial areas of the face.

Architecture Overview

- Four convolutional blocks
- Each block = Conv $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ **CBAM** $\rightarrow$ MaxPool
- Flatten $\rightarrow$ FC(512) $\rightarrow$ Dropout $\rightarrow$ Output layer

Training Setup

- Loss: Cross-Entropy
- Optimizer: Adam ($lr = 5e-4$)
- Epochs: 40

Result: 55.98 result, which represents a definite improvement since evidence of attention improvement was observed.

6.6 Transfer Learning EfficientNet-B0

EfficientNet-B0 proposes a more advanced structure integrating neural architecture search and compound scaling and thus is efficient and highly expressive. ImageNet pretrained weights are used to initialize the model and the last classification layer is an output head with seven classes to match the FER labels. Learned representations of features are frozen at earlier stages of the network and the top layers are fine-tuned with AdamW with a slower learning rate. A weighted cross entropy loss is also used to train the model in 40 epochs to counter weighted class disparity. Due to its well-trained feature extractor and balance scaling, EfficientNet-B0 delivers the best performance of all the models tested.

Modifications for FER

- Load the ImageNet-pretrained weights.
- Substitute final classifier by 7 class output layers.
- Freeze previous layers (feature extractor)
- Freeze and then just train only with AdamW ($lr = 1e-4$).
- Use weighted loss accuracy of cross-entropy.
- This model has good pretrained features and good scaling.
- Final accuracy: 66.49, the highest of all the models.

6.7 Evaluation and Interpretability

This section would determine the success of the project and the effect of the proposed project on the communities that the project operates.

The trained models are measured based on various quantitative and qualitative measures. The accuracy and confusion matrices indicate the performance of each emotion category in terms of recognition with EfficientNet-B0 performing better than most categories. The separability of classes further is supported by ROC and AUC scores: there are high scores on frequent expressions like happiness and neutral. Gradient-based saliency maps are computed to gain a better idea of how each individual prediction is made by the model by identifying the areas of the face that the model considers salient. The above visualizations show that the models have always given attention to the semantically significant features like the eyebrows, eyes and mouth, which justifies their interpretability and appropriateness in HCI applications.

1. Confusion Matrix

- Happy & neutral show highest separability
- Fear & disgust frequently confused due to subtle cues

2. ROC–AUC

- Most AUC values > 0.80 , indicating strong discriminative ability
- Low-sample class *contempt* exhibits more variability

3. Saliency Maps

- Happy → focus on mouth curvature
- Angry → eyebrows and upper eyelids
- Surprise → widened eyes, open mouth

Results and Evaluation

7.1 Overall Performance

In our experiments on the combined FER+CK+ test set:

Model	Preprocessing	Attention	Pretrained?	Test Accuracy (%)
Simple CNN	Face crop + CLAHE, 180×180	No	No	47.25
CBAM-CNN	Face crop + CLAHE, 180×180	CBAM	No	55.98
EfficientNet-B0	Face crop + CLAHE, 224×224	Implicit	ImageNet	66.49

Table 2. Comparison of model architectures and test accuracy on the combined FER-2013 + CK+ dataset.

Test accuracy of Simple CNN (baseline) is 47.25%.

CBAM-CNN also presents the accuracy of 55.98 in the last configuration we trained.

EfficientNet-B0 is the only model to achieve the highest test accuracy of 66.49%.

This proves that the performance increase by transfer learning of a robust ImageNet backbone provides a large performance improvement on FER+CK+.

7.2 Accuracy and Confusion based on the per-class.

In the case of EfficientNet-B0:

Happiness and classes of neutral performance are the best performers and have more frequency and distinctive forms. Anger and sadness do somewhat well.

The most difficult ones are fear and disgust, and they are often confused with each other and sadness.

These can be well observed in the confusion matrix of EfficientNet-B0 where misclassifications are noted to be concentrated around the negative emotions where facial cues are close (i.e., furrowed brows, tense mouth).

Class	Precision	Recall	F1-score	Support
angry	0.5765	0.5970	0.5865	985
disgust	0.6807	0.7687	0.7220	147
fear	0.5068	0.4658	0.4855	1039
happy	0.9101	0.8249	0.8654	1816
neutral	0.5295	0.5617	0.5451	1264
sad	0.7353	0.8672	0.7958	881
surprise	0.6144	0.5969	0.6055	1233
contempt	1.0000	1.0000	1.0000	11

Table 3. EfficientNet-B0 per-class precision, recall, and F1-score on the FER-2013 + CK+ test set.

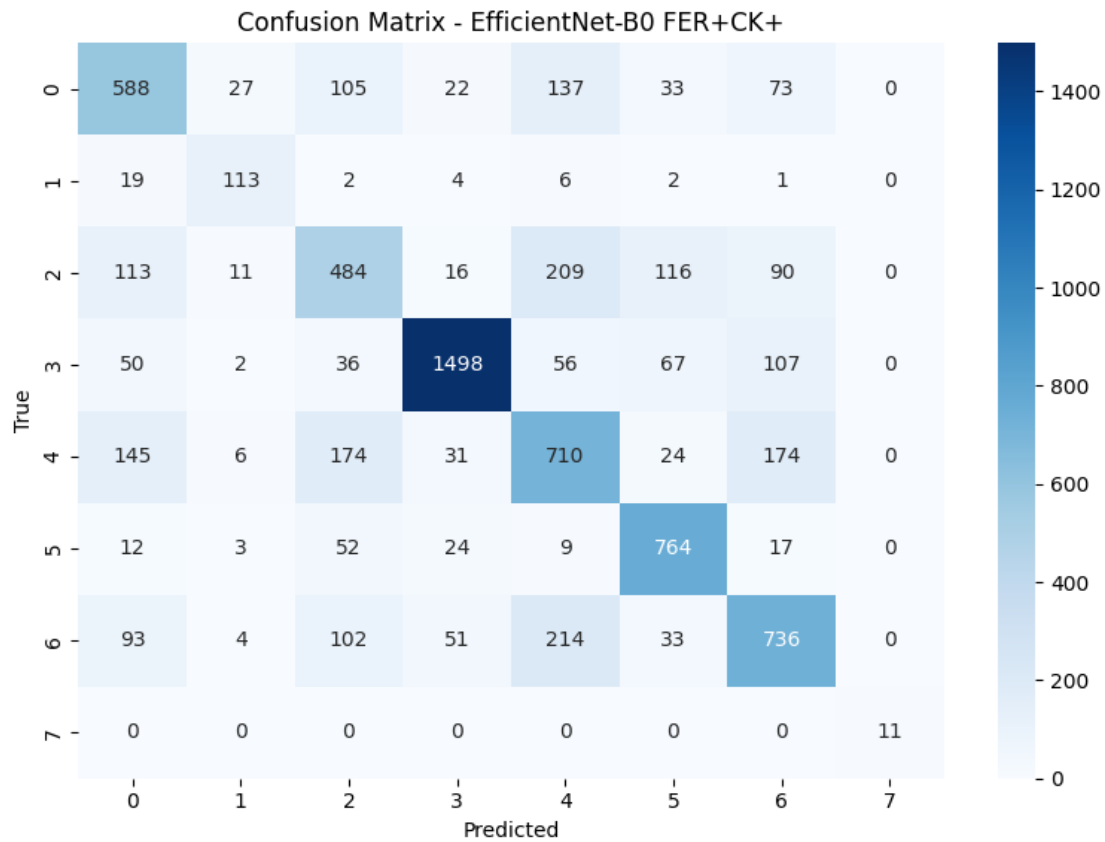


Figure 2. Confusion matrix for EfficientNet-B0 on the combined FER-2013 + CK+ test set.

7.3 ROC and AUC

To obtain ROC curve and AUC scores of each class, we compute:

The values of AUC in most classes are very high (grandiose than 0.8), which implies that they can be discriminated well even in the presence of varying thresholding.

Other low-frequency classes (e.g., contempt) have more varying curves because they have fewer positive examples.

The micro-averaged ROC curve provides a single connotation of the performance of all classes and gives a single AUC value that indicates the general quality of the overall ranking of the predictions.

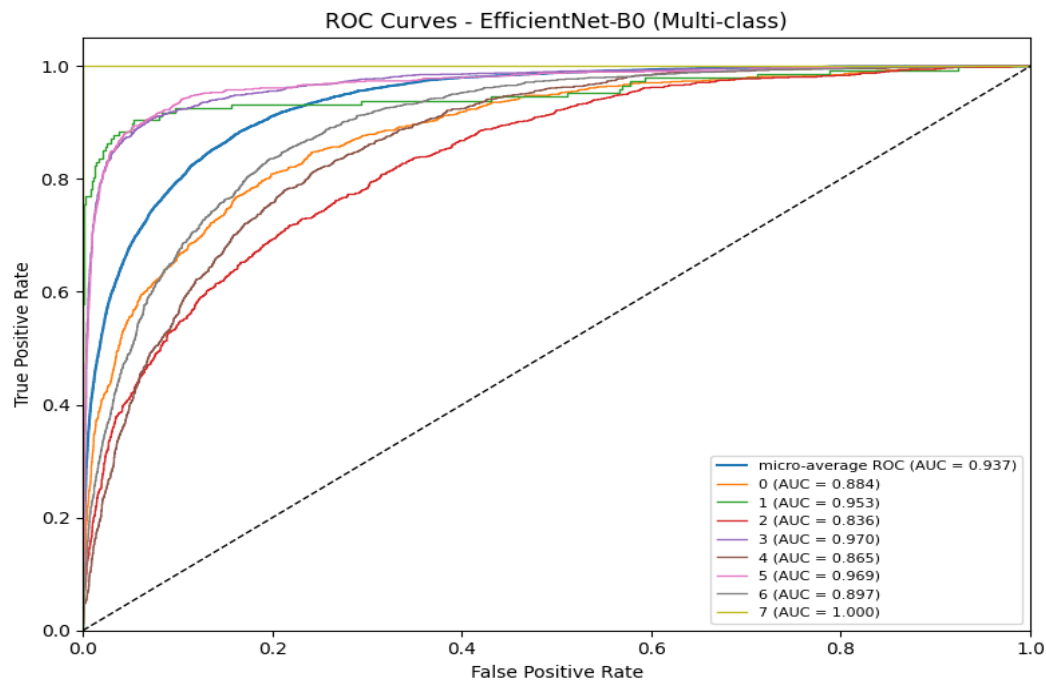


Figure 3. One-vs-rest ROC curves and micro-averaged ROC for EfficientNet-B0 on the combined FER-2013 + CK+ test set.

7.4 Interpretability Saliency

Our saliency maps are obtained as gradient-based and superimposed on face pictures:

In the case of happy the model pays attention to the parts of the mouth (curvature of smile) and the cheeks.

In case of anger, focus is drawn to eyebrows and upper eyelids (frowning).

To be surprising, the model puts an accent on large open eyes and an open mouth.

Such visualizations show that the model is fit to learn semantically meaningful cues, which are correlated with human intuition and Facial Action Units, as has been done previously in studies that associated CNN filters to AUS. Such interpretability can be useful in an HCI setting to debug misclassifications as well as instilling trust in a user. To give an example; suppose the system classifies a neutral face as sad, the designers can check the saliency map to check whether the shadows or occlusions are training the model to misread.

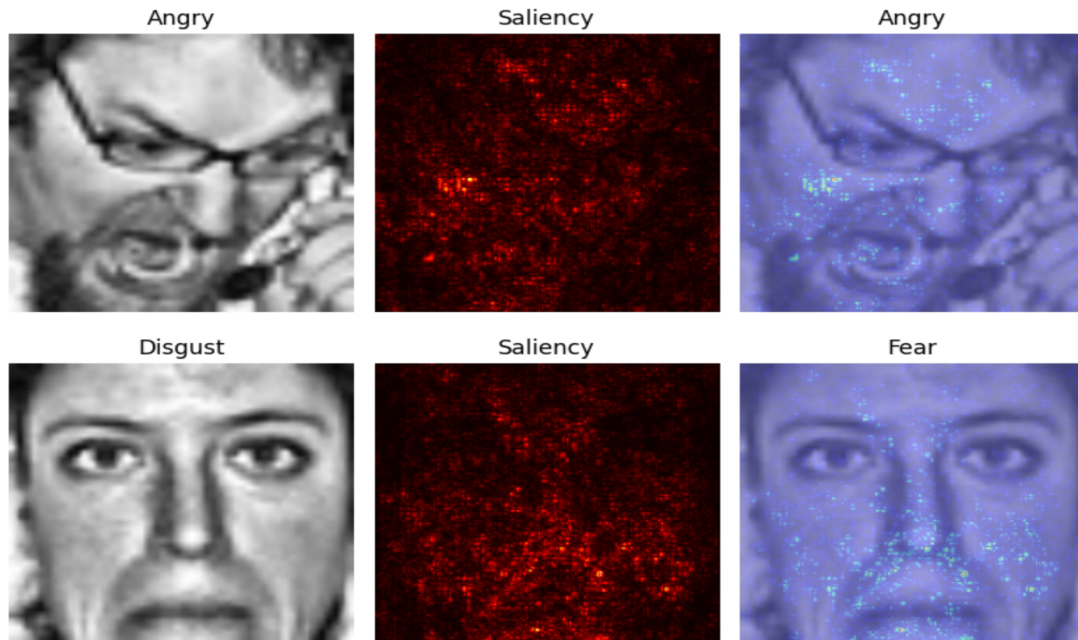


Fig 4.1 Gradient-based saliency map of Angry and Fear

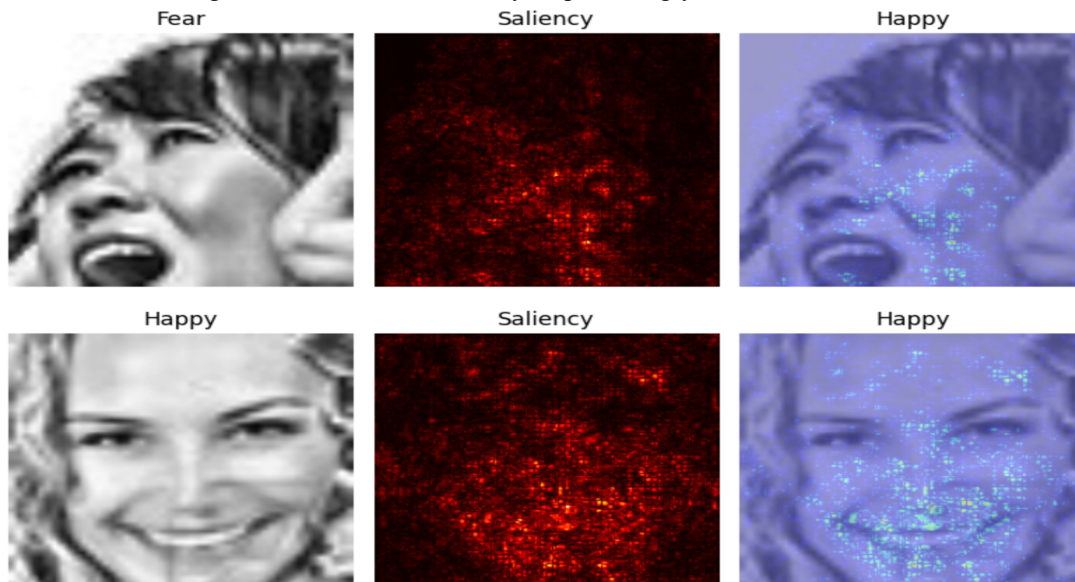


Fig 4.2 Gradient-based saliency map of Fear and Happy

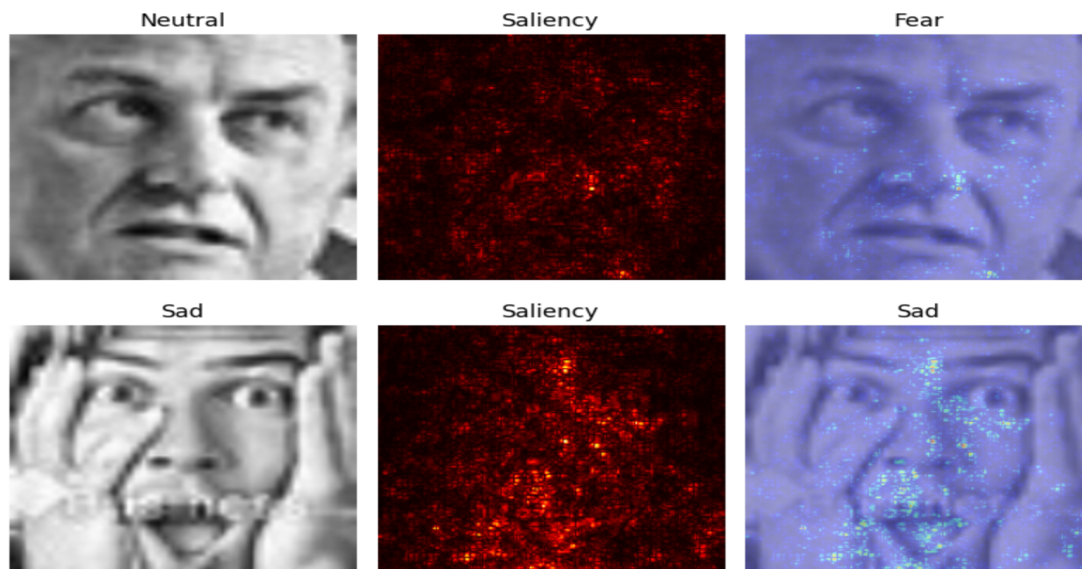


Fig 4.3 Gradient-based saliency map of Neutral and Sad

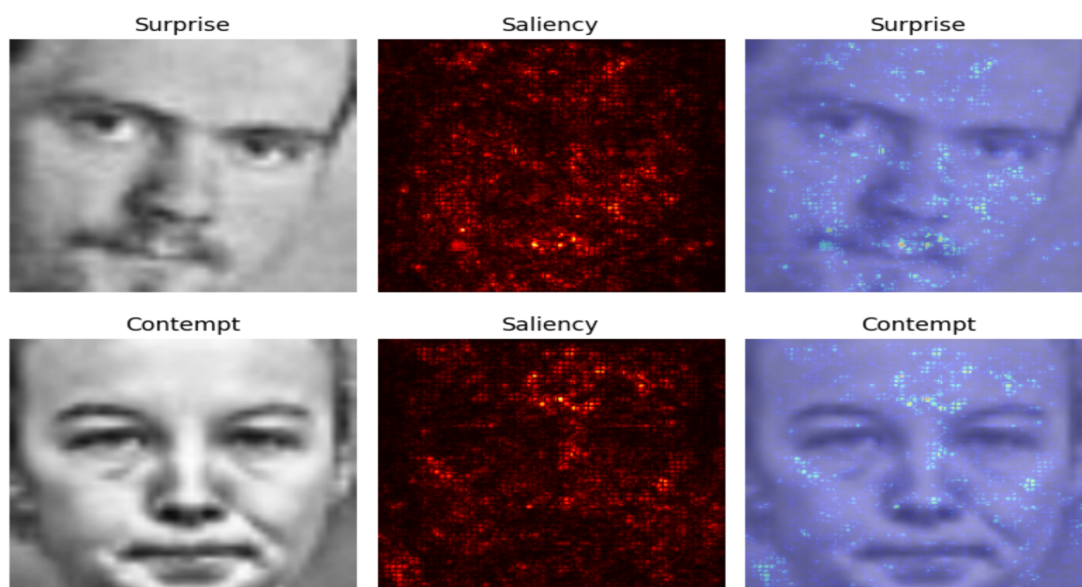


Fig 4.4 Gradient-based saliency map of Surprise and Contempt

7.5 Saliency Analysis of Correct and Misclassified predictions.

Gradient-based saliency maps were derived on correctly classified and misclassified samples to have a better comprehension of the way the model interprets facial features during prediction. In the cases of properly predicted emotions - Happy and Angry, the model always looked towards semantically significant portions of the body - the mouth curve in the case of happiness and brows and eyelids in the case of anger.

But in cases that were misclassified such as Neutral to Fear, the saliency highlights were less centralized and tended to form clumps of saliency around ambiguous shadowed areas or accidental facial features. This means that when the model puts too much emphasis on the noisy or misleading cues as opposed to the stable expression patterns, then there is high likelihood that the model will fail to produce accurate classifications. The value of such visual analysis is that it allows to comprehend the weaknesses of the model, enhance its interpretability, and focus the future optimization of preprocessing or architecture design.

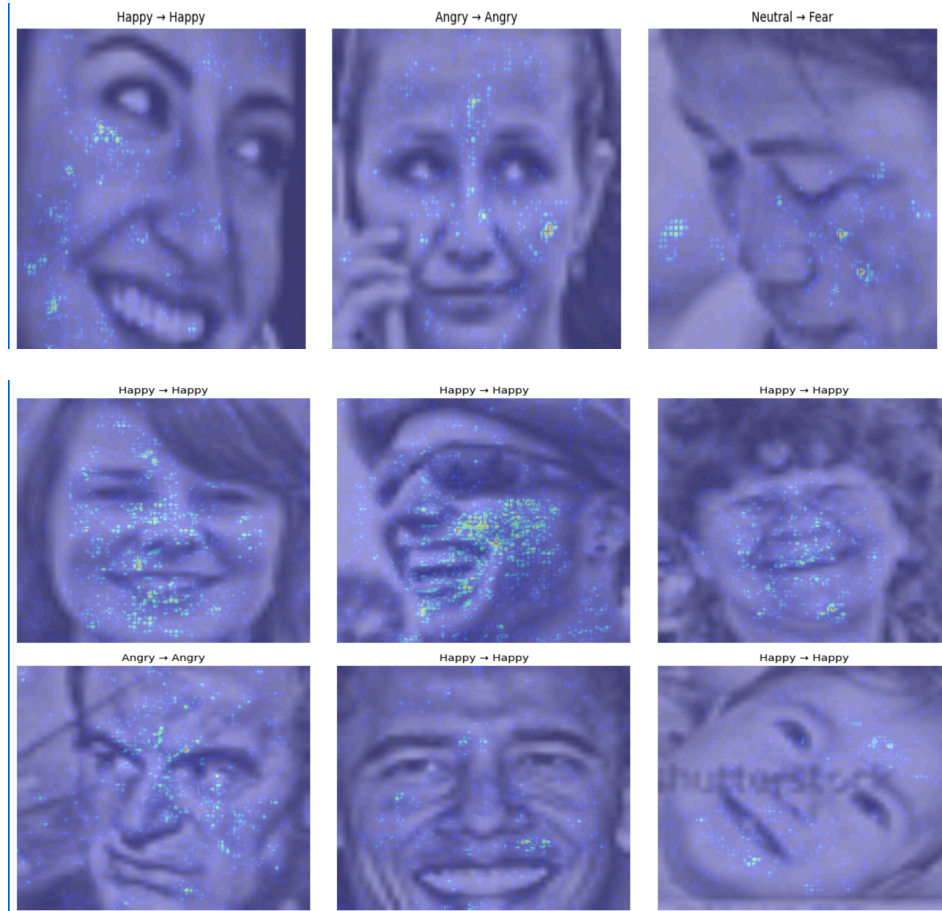


Fig 5 Gradient-based saliency overlays visualizing model focus for correctly predicted

7.6 Training and Validation Curves

The learning curves reflect how the EfficientNet-B0 model evolved during training across 40 epochs. The accuracy plot shows steady improvement in training accuracy, growing from approximately 83% to nearly 86%, while the validation accuracy stabilizes at approximately 67–68%, confirming consistent generalization without significant overfitting.

Similarly, it can be observed from the loss curves that the training loss decreases smoothly and remains low throughout, while the validation loss stays at approximately 1.0 with small fluctuations. This gap between the losses of training and validation further indicates that even though the model learns the patterns in the training dataset effectively, the inherent complexity and imbalance of FER-2013 and CK+ present challenges for validation performance. However, the stable behavior of validation proves that the model does not diverge and maintains reliable predictive capability across epochs.

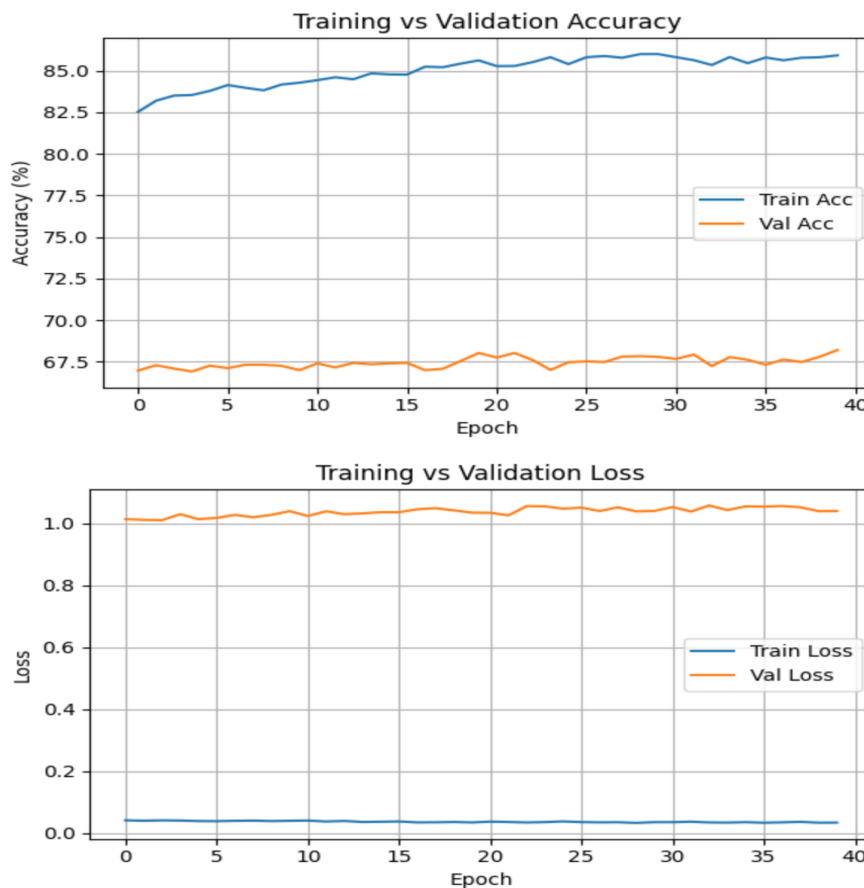


Fig 6 Training vs. Validation Accuracy and Loss curves for EfficientNet-B0

Discussion

8.1 Strengths

The proposed FER system has high performance whereby EfficientNet-B0 attains 66.49% accuracy on the combined FER-2013 + CK+ data set- competitive under course-level restrictions and dataset imbalance. The model is as well generally applicable to two visually dissimilar datasets, which advantage the use of single processing pipeline based on face-cropping and CLAHE. Moreover, the main advantage of the system is interpretability; confusion matrices, ROC curves, saliency visualizations are useful and provide valuable information on how the models arrive at the decisions. This openness is especially useful in an emotion-sensitive HCI system including adaptive tutoring, stress detection, and accessibility.

8.2 Limitations

Despite the encouraging findings, the system still has a few limitations. There are also emotions which cannot be easily categorized because of their low representation and small indications on the faces such as fear, disgust, and contempt. The datasets are skewed, affecting the accuracy per-class, and can be biased in the real-world application. Also, the existing variant does not use temporal information available in CK+ sequences and divides every picture separately; consideration of time-based models might enhance the detection of transitional expressions. Lastly is that the system uses faces alone, whereas in a real-life application of HCI, there is more than just facial usage that can be used to enhance the reliability of the system multimodal signals like voice or physiological information may be needed.

8.3 Novelty and Contribution

This project has a few contributions as compared to a standard implementation of a FER coursework. An integrated FER-2013 + CK+ preprocessing system was created, which allowed successful cross-dataset training by using uniform face-crop and contrast enhancement stages. The combination and the comparison of three architectures such as a custom CBAM-CNN and EfficientNet-B0 transfer learning provide information on attention mechanisms and scaling approaches to FER. Most significantly, the project focuses on interpretability by using saliency maps and ROC analysis, relating technical output to actual HCI applications.

Future Work

The enhancements to this FER-HCI system can be made in the future as:

Temporal Modeling: 3D CNNs, CNN+LSTM: Learning expression dynamics with time.

Multi-Task Learning: Reconstructing emotions and action units to enhance interpretability and robustness.

Fairness and bias reduction: Measuring performance in demographic groups and in underrepresented classes.

Dataset Enhancement: Growing or enhancing minority-emotion sample to decrease imbalance and increase generalization.

Real-Time Integration: Creation of adaptive interface based on live environment using a web camera to react to user emotions in real-time.

Model Efficiency: Training models (e.g., EfficientNet-Lite, quantization) that run on resource-limited and mobile devices.

These measures will contribute to the development of more trustworthy, more just and situational FER systems that can be applicable to practical HCI applications.

Conclusion

This project designed an end-to-end facial expression recognition (FER) pipeline with a goal of improving emotion-based human-computer interaction (HCI). Using CLAHE-based preprocessing and face-cropping with the FER-2013 and CK+ datasets, we trained and tested 3 deep-learning models including Simple CNN, CBAM-enhanced CNN and EfficientNet-B0. EfficientNet-B0 was the best performing one with the highest test accuracy of 66.49 and reflects the obvious effect of transfer learning and compound scaling on the FER performance.

We have found that the models classify the happiness and neutral expressions with high reliability indicating that there is high feature separation between these groups. Nevertheless, fear and disgust are still problematic because of the visual hints and lack of representation in datasets. We have viewed via confusion matrices, ROC curves, and saliency-map visualization the effect of attention mechanisms (CBAM) on focal attention to discriminative parts of the body, including the eyes, eyebrows, and mouth. These findings confirm that attention and modern CNN architecture are useful in interpretable FER.

Altogether, the presented piece of work shows that powerful, state-of-the-art CNN models can be used as powerful basis block of emotion-aware HCI systems, such as adaptive learning systems, mental-wellness systems, responsive interfaces. Although the outcomes are promising, the project also reveals the limitations associated with the imbalance of the datasets, the subtleties of the expressions, the issue of the demographic bias, and the problems with actual application. To develop workable, dependable, and ethically consistent FER systems, it is necessary to address these problems.

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